Project Name: Docker Swarm- Voting application

<table>
<tr><td>Objective</td><td>To deploy a scalable, multi-service voting application on a manager node, ensuring efficient orchestration, fault tolerance, and seamless monitoring through Docker visualizer</td></tr>
<tr><td>Problem Statement and Motivation</td><td>John, a DevOps engineer, is tasked with deploying a voting application through multiple microservices. By creating a Docker compose file and deploying it on a manager node in a distributed system, they ensure that each service is efficiently orchestrated and fault-tolerant.

To monitor the deployment, John integrates Docker visualizer as a microservice, providing real-time insights. This setup simplifies the deployment process, enhances scalability, and ensures the application runs smoothly in a production environment.</td></tr>
<tr><td>Create EC2 instances</td><td>
<table>
<tr><th>node</th><th>public-ip</th><th>private-ip</th></tr>
<tr><td>master</td><td>13.201.36.209</td><td>172.31.37.63</td></tr>
<tr><td>ndoe1</td><td>15.206.73.200</td><td>172.31.32.63</td></tr>
<tr><td>node2</td><td>13.127.206.197</td><td>172.31.42.250</td></tr>
</table>
<table>
<tr><th>Name</th><th>Instance ID</th><th>Instance state</th><th>Instance type</th><th>Status check</th><th>Alarm status</th><th>A</th></tr>
<tr><td>master</td><td>i-006c90bf918ff1854</td><td>Running</td><td>t3.medium</td><td>Initializing</td><td>View alarms +</td><td>a</td></tr>
<tr><td>node1</td><td>i-0d06144392b3d7c33</td><td>Running</td><td>t3.medium</td><td>Initializing</td><td>View alarms +</td><td>a</td></tr>
<tr><td>node2</td><td>i-0457caabe0e96bb0e</td><td>Running</td><td>t3.medium</td><td>Initializing</td><td>View alarms +</td><td>a</td></tr>
</table>
</td></tr>
<tr><td>Connect each server</td><td>ssh -i 002.pem ubuntu@13.201.36.209

<pre>Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-37-63:~$</pre></td></tr>
<tr><td>Install docker on all 3 instances</td><td>sudo apt update -y
sudo apt install docker.io -y
<pre>ubuntu@ip-172-31-37-63:~$ docker service status
docker: unknown command: docker service status

Usage: docker service

Run 'docker service --help' for more information
ubuntu@ip-172-31-37-63:~$ service docker status
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-03-21 07:49:51 UTC; 1min 28s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 1953 (dockerd)
      Tasks: 10
     Memory: 21.4M (peak: 22.0M)
        CPU: 327ms
     CGroup: /system.slice/docker.service
             └─1953 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock</pre></td></tr>
</table>

Install docker swarm on master node	# master: 172.31.37.63 docker swarm init --advertise-addr 172.31.37.63 ubuntu@ip-172-31-37-63:~$ sudo su root@ip-172-31-37-63:/home/ubuntu# docker swarm init --advertise-addr 172.31.37.63 Swarm initialized: current node (wlg52vm2jbp4null7pqnh8ea9) is now a manager. To add a worker to this swarm, run the following command: docker swarm join --token SWMTKN-1-465a6wdh46dd1gv0rjddysb4suqhzbutmo617wdwtbcgx3ggtw-d4mfi2ne6iq1x8ny9fojr3r6l 172.31.37.63:2377 To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions. root@ip-172-31-37-63:/home/ubuntu# #note down node join command
join node1 and node2 in swarm cluster	#node join command. Run below command on node1 and node2 docker swarm join --token SWMTKN-1-465a6wdh46dd1gv0rjddysb4suqhzbutmo617wdwtbcgx3ggtw-d4mfi2ne6iq1x8ny9fojr3r6l 172.31.37.63:2377 ubuntu@ip-172-31-42-250:~$ sudo su root@ip-172-31-42-250:/home/ubuntu# docker swarm join --token SWMTKN-1-465a6wdh46dd1gv0rjddysb4suqhzbutmo617wdwtbcgx3ggtw-d4mfi2ne6iq1x8ny9fojr3r6l 172.31.37.63:2377 This node joined a swarm as a worker. root@ip-172-31-42-250:/home/ubuntu#
Swarm cluster nodes	root@ip-172-31-37-63:/home/ubuntu# docker node ls ID HOSTNAME STATUS AVAILABILITY MANAGER STATUS ENGINE VERSION 3m06oxhsm8fytkb9977sdhd5w ip-172-31-32-63 Ready Active 28.2.2 wlg52vm2jbp4null7pqnh8ea9 * ip-172-31-37-63 Ready Active Leader 28.2.2 kieegz6vqm4dxusuk2kvce5fw ip-172-31-42-250 Ready Active 28.2.2 root@ip-172-31-37-63:/home/ubuntu#
overview of voting website	• Home page with 4 political party buttons • Vote page that saves the vote in a free MySQL database • Results page to see totals • Uses PHP + MySQL (works on free hosting like InfinityFree, 000webhost, or Hostinger free plans)
Structure:	voting-app/ ├── Dockerfile ├── index.php ├── vote.php ├── results.php └── db.php
Dockerfile	FROM php:8.2-apache RUN apt update -y RUN apt install git -y RUN git clone https://github.com/pradeepviswa/voting-app.git /var/www/html RUN ls /var/www/html RUN docker-php-ext-install mysqli #check mysqli RUN php -i | grep mysqli

Dockerfile	``` root@ip-172-31-37-63:/home/ubuntu# ls Dockerfile root@ip-172-31-37-63:/home/ubuntu# cat Dockerfile FROM php:8.2-apache RUN apt update -y RUN apt install git -y RUN git clone https://github.com/pradeepviswa/voting-app.git /var/ RUN ls /var/www/html RUN docker-php-ext-install mysqli #check mysqli RUN php -i | grep mysqli ```
Build Docker Image	docker build -t voting-image . ``` Step 7/7 : RUN php -i | grep mysqli ---> Running in a17be58c9b1b Additional .ini files parsed => /usr/local/etc/php/conf.d/docker-php-ext-my mysqli mysqli.allow_local_infile => Off => Off mysqli.allow_persistent => On => On mysqli.default_host => no value => no value mysqli.default_port => 3306 => 3306 mysqli.default_pw => no value => no value mysqli.default_socket => no value => no value mysqli.default_user => no value => no value mysqli.local_infile_directory => no value => no value mysqli.max_links => Unlimited => Unlimited mysqli.max_persistent => Unlimited => Unlimited mysqli.rollback_on_cached_plink => Off => Off API Extensions => mysqli ---> Removed intermediate container a17be58c9b1b ---> 9ecca2cd1bcf Successfully built 9ecca2cd1bcf Successfully tagged voting-image:latest root@ip-172-31-42-76:/home/ubuntu# ```
Create Overlay Network	docker network create \ --driver overlay \ voting-network ``` root@ip-172-31-37-63:/home/ubuntu# docker network create \ --driver overlay \ voting-network 0t9bomczex4x8sv1uyd5v5fhg root@ip-172-31-37-63:/home/ubuntu# ```
Check new network	docker network ls ``` root@ip-172-31-37-63:/home/ubuntu# docker network ls NETWORK ID NAME DRIVER SCOPE 08f51c0d1ecd bridge bridge local d925770e451c docker_gwbridge bridge local 3a9a2c4574c5 host host local ram0lhyud0t1 ingress overlay swarm 70531820823c none null local 0t9bomczex4x voting-network overlay swarm ```

Create MySQL Service	docker service create \
	--name voting-mysql \
	--network voting-network \
	-e MYSQL_ROOT_PASSWORD=root \
	-e MYSQL_DATABASE=voting_db \
	mysql:8
	`root@ip-172-31-37-63:/home/ubuntu# docker service create \`
	`--name voting-mysql \`
	`--network voting-network \`
	`-e MYSQL_ROOT_PASSWORD=root \`
	`-e MYSQL_DATABASE=voting_db \`
	`mysql:8`
	
	`gymjvc3dg70dzftv4zv8dxu0b`
	`overall progress: 1 out of 1 tasks`
	`1/1: running [==========>]`
	`verify: Service gymjvc3dg70dzftv4zv8dxu0b converged`
	`root@ip-172-31-37-63:/home/ubuntu#`

Create Web Service	docker service create \
	--name voting-web \
	--network voting-network \
	-p 8080:80 \
	voting-image
	`root@ip-172-31-37-63:/home/ubuntu# docker service create \`
	`--name voting-web \`
	`--network voting-network \`
	`-p 8080:80 \`
	`voting-image`
	`image voting-image:latest could not be accessed on a registry to record`
	`its digest. Each node will access voting-image:latest independently,`
	`possibly leading to different nodes running different`
	`versions of the image.`
	
	`dhlk952hqa1a9imdr0wu3iqz2`
	`overall progress: 1 out of 1 tasks`
	`1/1: running [==========>]`
	`verify: Service dhlk952hqa1a9imdr0wu3iqz2 converged`
	`root@ip-172-31-37-63:/home/ubuntu#`

Check service:	docker service ls
	`root@ip-172-31-37-63:/home/ubuntu# docker service ls`
	`ID NAME MODE REPLICAS IMAGE PORTS`
	`gymjvc3dg70d voting-mysql replicated 1/1 mysql:8`
	`dhlk952hqa1a voting-web replicated 1/1 voting-image:latest *:8080->80/tcp`

Docker containers	docker ps
	`root@ip-172-31-37-63:/home/ubuntu# docker ps`
	`CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES`
	`18d0c2bb861f voting-image:latest "docker-php-entrypoi…" About a minute ago Up About a minute 80/tcp voting-web.1.cm2171wc`
	`fd0eeb1dcb93 mysql:8 "docker-entrypoint.s…" 2 minutes ago Up 2 minutes 3306/tcp, 33060/tcp voting-mysql.1.071x63`

Container IDs	**18d0c2bb861f** voting-image:latest **fd0eeb1dcb93** mysql:8

Ping voting-mysql from voting-web

Connect voting image container

docker exec -it 18d0c2bb861f /bin/bash

```
root@ip-172-31-37-63:/home/ubuntu# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED
18d0c2bb861f   voting-image:latest   "docker-php-entrypoi…"   About a minute ago
fd0eeb1dcb93   mysql:8               "docker-entrypoint.s…"   2 minutes ago
root@ip-172-31-37-63:/home/ubuntu# docker exec -it 18d0c2bb861f /bin/bash
root@18d0c2bb861f:/var/www/html#
```

Install ping utility

apt install iputils-ping

```
root@18d0c2bb861f:/var/www/html# apt install iputils-ping
Installing:
  iputils-ping

Installing dependencies:
  linux-sysctl-defaults

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 57.0 kB
  Space needed: 211 kB / 2933 MB available

Continue? [Y/n] Y
```

ping voting-mysql

ping voting-mysql

```
root@18d0c2bb861f:/var/www/html# ping voting-mysql
PING voting-mysql (10.0.1.2) 56(84) bytes of data.
64 bytes from ip-10-0-1-2.ap-south-1.compute.internal (10.0.1.2): icmp_sec
64 bytes from ip-10-0-1-2.ap-south-1.compute.internal (10.0.1.2): icmp_sec
64 bytes from ip-10-0-1-2.ap-south-1.compute.internal (10.0.1.2): icmp_sec
64 bytes from ip-10-0-1-2.ap-south-1.compute.internal (10.0.1.2): icmp_sec
64 bytes from ip-10-0-1-2.ap-south-1.compute.internal (10.0.1.2): icmp_sec
```

Database Connection [info]

My **db.php** uses the service name as hostname:
$host = "voting-mysql";
Swarm DNS automatically resolves the service as shown above.

Enter MySQL Service to Create Table

get container ID:

docker ps
fd0eeb1dcb93 mysql:8

Connect mysql:

#password is root

docker exec -it **fd0eeb1dcb93** mysql -u root -p

```
root@ip-172-31-37-63:/home/ubuntu# docker exec -it fd0eeb1dcb93 mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.8 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Create table:

USE voting_db;

CREATE TABLE votes (
id INT AUTO_INCREMENT PRIMARY KEY,
party VARCHAR(50)
);

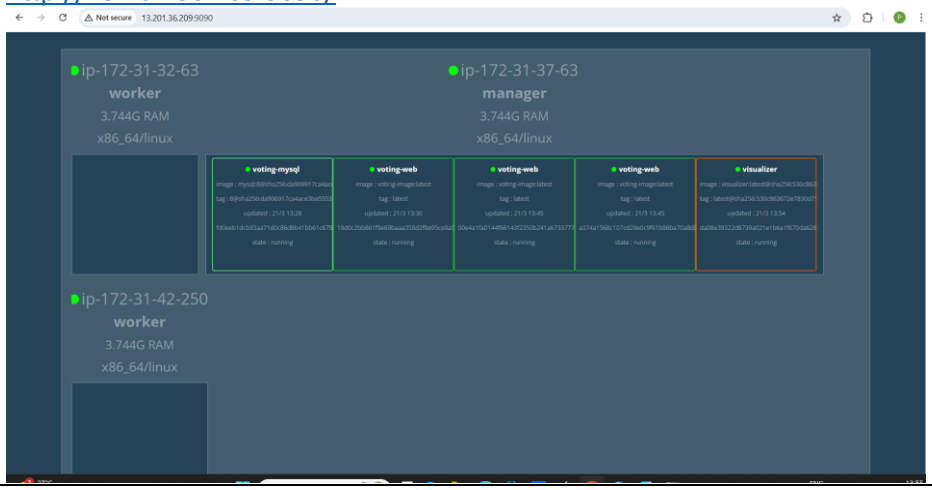
```
mysql> USE voting_db;
Database changed
mysql>
mysql> CREATE TABLE votes (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> party VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
```

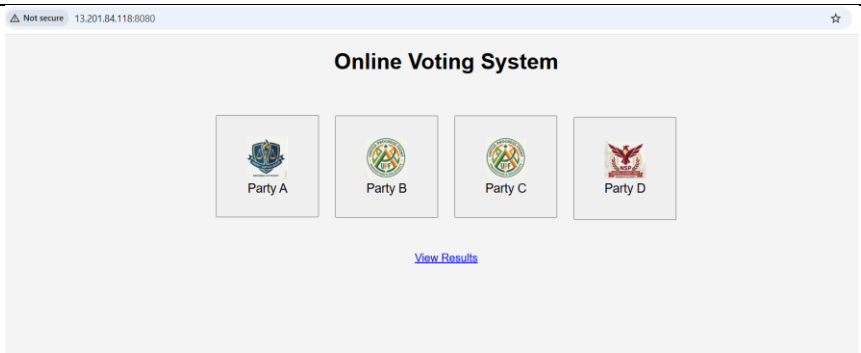
Scale the Web Service (Swarm advantage)	docker service scale voting-web=3
	`root@ip-172-31-37-63:/home/ubuntu# docker service scale voting-web=3`
	`voting-web scaled to 3`
	`overall progress: 3 out of 3 tasks`
	`1/3: running [==========>]`
	`2/3: running [==========>]`
	`3/3: running [==========>]`
	`verify: Service voting-web converged`
	`root@ip-172-31-37-63:/home/ubuntu#`

Access web stie	http://13.201.36.209:8080/
Docker Visualizer	Docker Visualizer does not communicate over container networks. It reads Docker data via: /var/run/docker.sock

Deploy Docker Visualizer as a service	docker service create \
	  --name=visualizer \
	  --publish=9090:8080 \
	  --constraint=node.role==manager \
	  --mount type=bind,src=/var/run/docker.sock,dst=/var/run/docker.sock \
	  dockersamples/visualizer
	`root@ip-172-31-37-63:/home/ubuntu# docker service create \`
	`  --name=visualizer \`
	`  --publish=9090:8080 \`
	`  --constraint=node.role==manager \`
	`  --mount type=bind,src=/var/run/docker.sock,dst=/var/run/docker.sock \`
	`  dockersamples/visualizer`
	`vxi5ur7uy2zt7udruyg4t4862`
	`overall progress: 1 out of 1 tasks`
	`1/1: running [==========>]`
	`verify: Service vxi5ur7uy2zt7udruyg4t4862 converged`
	`root@ip-172-31-37-63:/home/ubuntu#`

Access visualiser	http://13.201.36.209:9090/ 

Output:

Open Website And Click on different buttons multiple times	
When clicked on any button here is the output	
See Results page	

How voting app website is configured?

Git location	https://github.com/pradeepviswa/voting-app.git

db.php	<?php
	
	$host = "voting-mysql";
	$user = "root";
	$password = "root";
	$database = "voting_db";
	
	$conn = new mysqli($host,$user,$password,$database);
	
	if ($conn->connect_error) {
	 die("Connection failed: " . $conn->connect_error);
	}
	
	?>

index.html	<!DOCTYPE html> <html> <head> <title>Online Voting System</title> <link rel="stylesheet" href="style.css"> </head> <body> <h1>Online Voting System</h1> <form action="vote.php" method="POST"> <button name="party" value="Party A"> <img src="img/party1.png"> Party A </button> <button name="party" value="Party B"> <img src="img/party2.png"> Party B </button> <button name="party" value="Party C"> <img src="img/party3.png"> Party C </button> <button name="party" value="Party D"> <img src="img/party4.png"> Party D </button> </form> <a href="results.php">View Results</a> </body> </html>

results.php	<?php
	include "db.php";
	?>
	
	<!DOCTYPE html>
	<html>
	<head>
	<title>Voting Results</title>
	</head>
	
	<body>
	
	<h1>Voting Results</h1>
	
	<?php
	
	$parties = ["Party A","Party B","Party C","Party D"];
	
	foreach($parties as $p){
	
	$result = $conn->query("SELECT COUNT(*) as total FROM votes WHERE party='$p'");
	$row = $result->fetch_assoc();
	
	echo "<h3>$p : ".$row['total']." votes</h3>";
	
	}
	
	?>
	
	<a href="index.html">Back to Vote</a>
	
	</body>
	</html>

style.css	body{
	font-family: Arial;
	text-align:center;
	background:#f4f4f4;
	}
	
	.container{
	margin-top:50px;
	}
	
	button{
	width:150px;
	height:150px;
	margin:10px;
	font-size:18px;
	cursor:pointer;
	}
	
	img{
	width:60px;
	}

vote.php	<?php include "db.php"; if(isset($_POST['party'])){ $party = $_POST['party'];

	$sql = "INSERT INTO votes (party) VALUES ('$party')"; $conn->query($sql); echo "<h2>Thank you for voting!</h2>"; echo "<a href='results.php'>See Results</a>"; } ?>
Image files here	img/party1.png img/party2.png img/party3.png img/party4.png